Joseph Catanzarite

Artificial Intelligence Researcher | Data Scientist | Machine Learning & Astrophysics

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jcatanza.github.io | Google Scholar

PROFESSIONAL SUMMARY

Artificial intelligence (AI) researcher and data scientist with experience applying machine learning, Bayesian statistics, and scientific computing to National Aeronautics and Space Administration (NASA) missions and teaching. Built a Random Forest classifier (97% accuracy) for Kepler exoplanet candidate vetting, published in NASA's Kepler Data Release 24 (DR24) catalog; led the Jet Propulsion Laboratory (JPL) analysis team in a five-team double-blind exoplanet detection test. Completing an M.S. in Artificial Intelligence at Johns Hopkins (expected 2028) with work in large language models (LLMs), generative AI, and AI agents. Author of peer-reviewed papers (~5,900 citations, h-index 32). Instructor and research mentor in AI, data science, physics, and astronomy.

CORE SKILLS

AI & Machine Learning: Large Language Models (LLMs), Generative AI, AI Agents, Deep Learning, Convolutional Neural Networks (CNNs), Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Natural Language Processing (NLP), Computer Vision, Supervised Learning, Random Forest, k-Nearest Neighbors, Model Training & Evaluation, Large Language Model Operations (LLMOps), Claude (Anthropic)

Statistics & Modeling: Bayesian Modeling & Inference, Statistical Inference, Hierarchical Modeling, Monte Carlo Simulation, Time Series Analysis, Signal Processing, Wavelet Analysis, Experimental Design, Causal Inference, Optimization

Scientific Computing & Astrophysics: Exoplanet Detection, Astrometry, Radial Velocity, Transit Photometry, Astrophysical Data Analysis, Numerical Simulation, Data Validation, Pipeline Development, High-Performance Computing

Programming & Tools: Python, MATLAB, Structured Query Language (SQL), PyTorch, TensorFlow, Keras, fastai, scikit-learn, NumPy, pandas, Jupyter, Git/GitHub, Subversion (SVN), Google Cloud Platform

PROFESSIONAL EXPERIENCE

AI and Data Science Mentor (Learning Coach) | Elton University / *Contract, Remote* *Oct 2026 – Present*

- Supervise and guide hands-on student projects and education in artificial intelligence, machine learning, and data science.

LLM Researcher, Handshake AI Fellow | Handshake AI / *Freelance, Remote* *Sep 2026 – Present*

- Conduct research and evaluation work on AI agents and large language models (LLMs).

Research Mentor in AI, Astronomy, and Physics | Nova Scholar Education / *Part-time, Remote* *Apr 2026 – Present*

- Mentor young researchers one-on-one on astrophysics and machine learning projects that lead to journal publications, developing their research, critical thinking, and scientific communication skills.

Career Break | Family and well-being; preparation for AI graduate school *Jan 2022 – Dec 2025*

Data Scientist / Data Science Instructor | BloomTech / *Full-time, Remote* *Aug 2021 – Dec 2021*

- Developed and taught a Data Science and Machine Learning curriculum for an online Bachelor's degree program; supervised student projects and work-study students.

Data Scientist / Lead Data Science Instructor | Make School / *Full-time, Remote* *Nov 2020 – Aug 2021*

- Developed and delivered Deep Learning & Computer Vision, Machine Learning in Production, and Intro to Data Science courses; led the capstone Data Science Technical Seminar (research to deployment).

Volunteer & Fellowship Roles in AI | DataKind, TWiML & AI, University of San Francisco *Oct 2018 – Apr 2021*

- **Volunteer Data Scientist, DataKind Medic Mobile (2019–2020):** improved models for maternal and child health outcomes.
- **Volunteer AI Study Group Lead, TWiML & AI (2019–2021):** led remote study groups on machine learning, NLP (CS224N), and deep learning math.
- **Fast.ai International Fellow, University of San Francisco (2018–2020):** organized study groups and developed fast.ai deep learning and NLP curriculum.

Data Scientist | Search for Extraterrestrial Intelligence (SETI) Institute / *Mountain View, CA* *Jan 2013 – Aug 2017*

- As astronomer and scientific programmer attached to NASA's Kepler Mission Science Office, built a Random Forest "autovetter" that classifies Kepler exoplanet candidates with 97% accuracy; published in the DR24 catalog.
- Led synthetic flux-level transit injection experiments for Kepler's Occurrence Rate Data Products (delivered to NASA Headquarters); applied k-Nearest Neighbors (kNN) to estimate detection efficiency versus signal-to-noise ratio, cutting NASA Pleiades supercomputer time by 85%.
- Created Kepler Data Processing Pipeline algorithms in MATLAB under SVN: wavelet-based denoiser (Presearch Data Conditioning), ghost diagnostic and ephemeris matching (Data Validation), thruster firing detector (Photometric Analysis).

Project Staff Scientist / Research Technologist IV | NASA Jet Propulsion Laboratory (JPL) / *Pasadena, CA* *Jan 2003 – Jan 2013*

- As an astronomer on the Space Interferometry Mission (SIM) planet-finding team, led JPL's Analysis Team in NASA's double-blind test, coordinating five teams to reach consensus solutions from joint astrometric and radial velocity data.
- Invented a novel hierarchical algorithm (nonlinear orbit fitting) to detect exoplanets and characterize their masses and orbits; played a key role in demonstrating SIM's capacity to detect Earth-mass planets.
- Ran mission-level Monte Carlo simulations of exoplanet-finding capability (astrometric, radial velocity, coronagraphic) and starspot effects; devised wavefront-sensing and line-of-sight jitter algorithms for the Advanced Mirror Development Testbed.

Earlier Experience

- **Senior Engineer and Senior Application Developer, California Institute of Technology:** astrometric trade studies; SIM Micro-Arcsecond Metrology testbed analysis; Wide Field Infrared Explorer (WIRE) star-tracker data.
- **Professor of Physics and Astronomy, Cypress College:** chaired the Physical Sciences Department; developed and taught physics and astronomy courses (lecture and laboratory).

SELECTED AI PROJECTS

Professor.Claude.AI — autonomous AI research agent | github.com/jcatanza/professor-claud-ai *May 2026 – Present*

- Python/LLMOps pipeline that nightly ingests new arXiv papers, triages them, and deep-reads the best with Claude into structured analyses; persistent semantic memory; daily email and GitHub digest.

Chinese Character Recognition and Generation — Johns Hopkins *2026*

- Compared conditional VAE, conditional GAN, and hybrid VAE-GAN (C-VAEGAN) generative data augmentation on Chinese-MNIST, scored with Fréchet Inception Distance (FID), Structural Similarity Index (SSIM), Learned Perceptual Image Patch Similarity (LPIPS), and downstream CNN accuracy; designed a novel two-stage quality filter for synthetic samples.

Shots, Not Stars! 2026 World Cup Prediction Model | View project page *Jul 2026*

- Random Forest and bootstrap sampling on 17 team-level statistics from 67 Fédération Internationale de Football Association (FIFA) World Cup matches; 62% win probability for Spain closely matched Polymarket.

A Primer on Modal and Epistemic Logic | jcatanza.github.io/logic *2026*

- Self-contained tutorial from propositional logic to modal and epistemic proofs about Belief-Desire-Intention (BDI) agents.

Science & Machine Intelligence — LinkedIn newsletter | Subscribe on LinkedIn *Monthly*

- Monthly newsletter (203 subscribers) on current boundaries and future trends in AI and scientific discovery.

PUBLICATIONS

~5,900 citations | h-index 32 | i10-index 52 — full list on Google Scholar

- **First author:** "Catching Up with the Fastest Star in the Galaxy: A Two-Passage GRAVITY+ Campaign to Detect S301's Schwarzschild Precession," arXiv:2609.06936 (2026), submitted to Publications of the Astronomical Society of the Pacific (PASP); Catanzarite & Shao, "The Occurrence Rate of Earth Analog Planets Orbiting Sun-like Stars," The Astrophysical Journal (ApJ) 738, 151 (2011); "Astrometric Detection of Terrestrial Planets in the Habitable Zones of Nearby Stars with SIM PlanetQuest," PASP 118, 1319 (2006).
- **Kepler team (selected):** PASP 126 (2014), wavelet error correction; ApJ 809 (2015), planet occurrence rates; The Astronomical Journal (AJ) 150 (2015), Kepler-452b; PASP 130 (2018), Data Validation; AJ 161 (2021), rocky habitable-zone planets.

EDUCATION

M.S., Artificial Intelligence | Johns Hopkins University, Whiting School of Engineering *Jan 2026 – May 2028 (expected)*
M.S., Electrical Engineering | University of Southern California (Rockwell Fellowship)
M.S., Physics | University of Arizona (Mathematics minor)
B.S., Mathematics | California Institute of Technology (Caltech)

CERTIFICATIONS & TRAINING

Deep learning & machine learning: Practical Deep Learning for Coders 1 & 2 (fast.ai, University of San Francisco, 2018); Deep Learning Specialization (deeplearning.ai, 2018); Machine Learning Specialization (University of Washington, 2017); Foundations of Statistical Machine Learning (Arizona State University, 2018); TensorFlow 2 (Imperial College London, 2021).
Statistics, math & data: Learning from Data (Caltech); Convex Optimization (Stanford); Causal Inference (University of Pennsylvania); SQL for Data Science (University of California, Davis); Python for Research (HarvardX).

HONORS & AWARDS

- NASA JPL Group Achievement Awards (5): SIM Science Office Team (2011); SIM Planet Finding Capability Study (2010); SIM Planet Hunter Mission Study (2007); SIM Microarcsecond Metrology Testbed (2003, 2005)
- NASA/University Joint Venture (JOVE) Research Fellowship; NASA Summer Faculty Fellowship; National Institute for Staff and Organizational Development (NISOD) Excellence Award; Cypress College Faculty Development Award